Retail Industry

Knowledge Guide for Data Science Freshers

Brick-and-Mortar | E-Commerce | Omnichannel | Merchandising | Supply Chain

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About This Guide

This guide helps new data science team members quickly understand the Retail industry. It covers the retail landscape, store operations, merchandising, e-commerce, supply chain, customer analytics, and the key terminologies used by retail clients. Whether your client runs physical stores, an online marketplace, or a combined omnichannel business, this guide prepares you to contribute from day one.

How to use: Read through each section in order. Terminology tables and KPI sections are especially important, as retail clients use these terms constantly in reviews, dashboards, and project briefs.

INTERNAL TRAINING DOCUMENT

Section 1: What is Retail?

Retail is the process of selling goods or services directly to end consumers through various channels — physical stores, online platforms, catalogues, or a combination. It is one of the largest sectors globally and one of the most data-intensive, generating enormous volumes of transaction, inventory, customer, and behavioral data every day.

1.1 Defining Characteristics of Retail

- Consumer-facing: The retailer is the final link between the supply chain and the end customer.
- High transaction volume: Millions of purchases daily across thousands of stores or digital touchpoints.
- Thin margins: Retail operates on low profit margins per unit — efficiency and volume are everything.
- Highly competitive: Price, convenience, product range, and experience all determine which retailer a consumer chooses.
- Seasonality-driven: Retail demand spikes during festivals, holidays, and sale events (Diwali, Christmas, Big Billion Day).
- Real-time decisions: Pricing, promotions, and stock replenishment often need to happen in near-real-time.

1.2 Types of Retail Formats

Retail Format	Description	Indian Examples
Hypermarket	Very large stores (50,000+ sq ft) selling food, clothing, electronics, and general merchandise under one roof.	Big Bazaar, Star Bazaar
Supermarket	Large grocery-focused stores (5,000–50,000 sq ft) with full range of food and household products.	DMart, Reliance Smart, Spencer's
Convenience Store	Small stores (under 2,000 sq ft) focused on quick purchases, high footfall locations.	7-Eleven, Reliance Fresh
Department Store	Multi-category stores focused on apparel, home, and lifestyle. Organized by department.	Shoppers Stop, Lifestyle, Pantaloons

Specialty Store	Stores focused on a single product category — electronics, footwear, books, sports.	Croma, Titan, Crossword, Decathlon
Discount / Value Retailer	Focus on everyday low pricing with minimal frills. High volumes, low cost.	DMart, V-Mart, Vishal Mega Mart
Quick Commerce / Dark Store	Ultrafast delivery (10–30 mins) from small fulfillment centers. No customer walk-ins.	Blinkit, Zepto, Swiggy Instamart
E-Commerce Marketplace	Online platform where multiple sellers list products. Retailer earns commissions.	Amazon, Flipkart, Meesho, Myntra
D2C (Direct to Consumer)	Brand sells directly to consumers online without middlemen.	Lenskart, Mamaearth, boAt, WOW Skin
Franchise Retail	Standardized retail format run by individual franchise owners under a brand license.	McDonald's, Subway, Fabindia

Key Takeaway

Every retail data science project must understand which format the client operates in. A hypermarket's challenges (assortment breadth, footfall analytics, store-level forecasting) are very different from a D2C brand's challenges (acquisition, retention, personalization, return rate management).

Section 2: Store Operations — The Core of Physical Retail

For brick-and-mortar retailers, store operations are central to every data science engagement. You need to understand how a store runs to interpret the data it generates.

2.1 Store Structure & Operations Terminology

Term	Meaning
Store Manager / SM	Person responsible for the overall performance of a store — sales, staff, customer experience, and compliance.

Floor Area / Selling Area	Total floor space of the store. Selling area = space used to display products and serve customers. Back-room/stockroom is excluded.
GLA (Gross Leasable Area)	Total area of a retail property available for lease to tenants. Used in mall and property management contexts.
Footfall / Traffic	Number of customers (or people) who enter a store in a given time period. A fundamental input to conversion rate calculations.
Conversion Rate	Percentage of store visitors who make a purchase. Conversion Rate = Transactions / Footfall x 100.
Basket Size / Average Transaction Value (ATV)	Average amount spent per customer per visit. ATV = Total Sales / Number of Transactions.
UPT (Units per Transaction)	Average number of items bought in a single transaction. A measure of customer basket depth.
Dwell Time	Average time a customer spends in the store. Longer dwell time is associated with higher spend.
Shrinkage	Inventory loss from theft (shoplifting, employee theft), administrative errors, or damage. Typically 1-3% of sales.
Store Clustering	Grouping stores by performance, demographics, size, or format to inform tailored strategies. A common DS project.
Same-Store Sales (SSS) / Comparable Store Sales	Sales growth from stores open for at least 12 months. Removes the effect of new store openings to show organic growth.
New Store Ramp-up	Period after a store opens when sales gradually build toward steady-state. Important for forecasting new store performance.
Planogram	A visual diagram prescribing exactly where each product should be placed on shelves. Drives visibility and impulse purchase.
Gondola / End Cap	Gondola = freestanding shelf unit on the shop floor. End Cap = display at the end of an aisle, highest visibility spot.
Hot Zone / Cold Zone	Hot zones = high-traffic areas of the store with maximum visibility. Cold zones = low-traffic corners. Product placement strategy uses this.
Shrink / Write-off	Shrink = inventory loss. Write-off = accounting removal of stock that cannot be sold (expired, damaged, stolen).
Stockroom / Back Office	Storage area behind the sales floor where additional inventory is held. Stockroom to floor replenishment speed affects shelf availability.

2.2 Common Data Science Use Cases in Store Operations

- Footfall Analytics: Predict store traffic patterns to optimize staffing and store layout.
- Conversion Rate Optimization: Identify which stores have low conversion despite high footfall and diagnose causes.
- Store Clustering: Group stores by profile to create targeted assortment, pricing, and promotional strategies.
- New Store Performance Prediction: Forecast a new store's ramp-up trajectory using catchment area demographics and analogous stores.
- Shrinkage Detection: Build anomaly detection models on inventory data to identify stores with unusual shrinkage patterns.
- Staff Scheduling Optimization: Use footfall and transaction forecasts to optimize staff levels across shifts and days.

Section 3: Merchandising & Inventory Management

Merchandising is about having the right product, at the right place, at the right time, at the right price, in the right quantity. It is one of the most analytically rich areas in retail, and where data scientists spend much of their time.

3.1 Merchandising Terminology

Term	Meaning
Assortment	The selection of products a retailer carries. Width = number of different categories. Depth = number of variants within a category.
Assortment Planning	Deciding which SKUs to carry in which stores, in what quantities, for a given season or period. Key DS use case.
Localization	Customizing the product assortment and promotions for a store based on local customer demographics and preferences.
Category Management	The process of managing a product category as a strategic business unit — assortment, placement, pricing, promotion.
Category Captain	A supplier appointed by the retailer to provide category insights and recommendations. E.g., P&G may be category captain for hair care at a supermarket.

Private Label / Own Brand	Products manufactured by a third party but sold under the retailer's own brand name. Higher margins for the retailer. E.g., DMart's D'Lite brand.
National Brand	Products manufactured and marketed under the manufacturer's own brand. E.g., Nestle KitKat, Nike shoes.
SKU Rationalization	Reducing the number of SKUs carried to remove slow-movers and simplify operations. Often a DS-driven exercise.
Range Review	Periodic process of evaluating which products to keep, add, or drop from the assortment. Usually seasonal.
Open-to-Buy (OTB)	Budget available for purchasing additional inventory within a period. OTB = Planned Sales + Planned Ending Inventory - Beginning Inventory - On Order.
Markdown / Clearance	Reducing the price of a product to clear slow-moving or end-of-season stock. Markdown optimization is a key DS use case.
Mark-up / IMU (Initial Mark-Up)	The percentage added to cost price to arrive at the selling price. IMU = (Retail Price - Cost) / Retail Price.
Gross Margin Return on Investment (GMROI)	Gross Profit / Average Inventory Cost. Measures how much profit is earned for every rupee invested in inventory.
Planogram Compliance	Degree to which actual shelf placement matches the prescribed planogram. Non-compliance leads to lost sales.
Space Productivity	Sales per square foot of selling space. A standard metric to compare store or category performance.
Sell-Through Rate	Units Sold / Units Received x 100. How quickly inventory is selling. Low sell-through = markdown risk. Used heavily in fashion.
Weeks of Supply (WOS)	Current Inventory / Average Weekly Sales. How many weeks current stock will last at current rate of sale.
Dead Stock / Slow Mover	Products that have not sold for an extended period. Tied-up capital and space. Needs markdown or removal.
In-Stock Rate / On-Shelf Availability (OSA)	Percentage of time a product is available on the shelf when a customer wants it. Direct link to lost sales.
Replenishment	Restocking the shelves or warehouse when inventory falls below a threshold. Can be manual, automated, or ML-driven.

3.2 Inventory & Supply Chain Terminology

Term	Meaning
DC (Distribution Center)	Retailer's own warehouse that receives goods from suppliers and distributes to stores. Some retailers use vendor-managed DCs.
Cross-Docking	Goods arriving at the DC are immediately transferred to outbound trucks for delivery to stores, with minimal storage time.
Vendor Managed Inventory (VMI)	Supplier takes responsibility for monitoring and replenishing the retailer's stock. Reduces retailer's forecasting burden.
Lead Time	Time from placing a purchase order with a supplier to receiving the goods in the DC or store.
Safety Stock	Buffer inventory held to protect against demand variability or supply delays.
EOQ (Economic Order Quantity)	The optimal order quantity that minimizes total ordering and holding costs.
Reorder Point	Inventory level at which a new purchase order is triggered. = Lead Time Demand + Safety Stock.
Inventory Turnover	Net Sales / Average Inventory. How many times inventory is sold and replaced in a period.
GMROI	Gross Margin / Average Inventory at Cost. Profitability return per rupee of inventory investment.
Aging Inventory	Inventory that has been in stock for longer than a defined period. Risk of markdowns or write-offs.
Purchase Order (PO)	Formal document sent to a supplier authorizing purchase of specified goods at an agreed price.
ASN (Advance Shipment Notice)	Notification from supplier to retailer about an incoming shipment before it arrives at the DC.
Receiving / GRN (Goods Received Note)	Process of accepting and logging incoming inventory at the DC or store. GRN is the document confirming receipt.
Shrinkage	Inventory loss due to theft, damage, expiry, or administrative error. Tracked as % of sales.
FIFO / FEFO	FIFO = First In First Out (dispatch in receipt order). FEFO = First Expired First Out (dispatch nearest expiry first).

Section 4: Pricing & Promotions

Pricing and promotions are the most visible levers a retailer has. Getting them right requires sophisticated data science — from price elasticity modeling to promotion response prediction.

4.1 Pricing Terminology

Term	Meaning
EDLP (Everyday Low Pricing)	Strategy of maintaining consistently low prices rather than running frequent temporary promotions. Associated with DMart, Walmart.
Hi-Lo Pricing	Strategy of setting high regular prices but offering frequent, deep promotions. Common in department stores and grocery.
Keystone Pricing	Retail price = 2x the wholesale/cost price. A simple mark-up rule of thumb.
Price Elasticity	Measure of how sensitive demand is to a price change. Elastic = demand changes a lot. Inelastic = demand barely changes. Critical input to pricing decisions.
Cross-Price Elasticity	How a price change on one product affects demand for another. Positive = substitutes (raise A's price, sell more B). Negative = complements (raise A's price, sell less B).
Zone Pricing	Charging different prices in different geographic zones or store clusters based on local competition and demand.
Dynamic Pricing	Adjusting prices in real time based on demand, competition, inventory levels, or time of day. Common in e-commerce, airlines, hotels.
Competitive Pricing	Setting prices based on what competitors charge. Retailers track competitor prices using web scraping and price intelligence tools.
Price Architecture	The range of price points offered within a category from budget to premium, ensuring the retailer captures all shopper segments.
Bundle Pricing	Selling multiple products together at a combined price lower than the sum of individual prices. E.g., Buy shampoo + conditioner bundle.
Loss Leader	A product priced at or below cost to drive store traffic, with the expectation that customers buy other profitable items.

Anchor Pricing	Displaying a high original price crossed out next to a lower 'sale' price to make the deal appear more attractive.
Price Index / Price Position	Retailer's prices vs. competitors, expressed as an index. Price Index of 98 means 2% cheaper than competition on average.

4.2 Promotions Terminology

Term	Meaning
TPR (Temporary Price Reduction)	A time-limited price reduction on a product. The most common promotional mechanic.
Promotional Lift	Incremental sales volume during a promotion above the baseline (what would have sold without the promotion).
Baseline Sales	Expected sales in the absence of a promotion. Separated from actual sales using statistical models.
Promotional ROI	Incremental profit generated by a promotion / Promotional cost. If below 1, the promotion lost money.
Cannibalization	When a promoted product takes sales away from another product in the same category or from the same retailer.
Halo Effect	When promoting a product lifts sales of related, non-promoted products in the same category.
Forward Buying / Pantry Loading	Customers buying more than they need during a promotion to stock up at the discounted price. Suppresses future demand.
Feature	Product being prominently displayed in a promotional flyer, website banner, or in-store signage.
Display	A special in-store placement beyond the normal shelf — end cap, freestanding display unit, floor display.
APC (Average Promotional Cost)	Total promotional investment / Units sold during the promotion.
Coupon	Voucher entitling the holder to a discount. Can be paper, digital, or app-based.
Loyalty Points / Cashback	Reward earned on purchase, redeemable on future purchases. Drives loyalty and repeat visits.

Flash Sale	Very short-duration (hours to 1–2 days) sale with very deep discounts. Common in e-commerce. Creates urgency.
Seasonal Sale	Major promotional events tied to seasons or occasions: End-of-Season Sale (EOSS), Diwali Sale, Republic Day Sale.
Post-Promotion Dip	Drop in sales immediately after a promotion ends, as customers who forward-bought don't need to purchase again.

4.3 Common Data Science Use Cases in Pricing & Promotions

- Price Elasticity Modeling: Quantify how demand responds to price changes at SKU and category level.
- Promotion Response Modeling: Predict sales lift and ROI for planned promotions before they are executed.
- Markdown Optimization: Determine optimal timing and depth of markdowns to clear slow-moving stock while maximizing revenue.
- Competitive Price Monitoring: Scrape and analyze competitor prices to inform dynamic pricing decisions.
- Personalized Promotions: Target the right offer to the right customer based on their purchase history and preferences.

Section 5: E-Commerce & Omnichannel Retail

E-commerce has transformed retail. Most large retailers now operate across physical stores and digital channels simultaneously — this is called omnichannel retail. Data scientists working with retail clients must understand both worlds.

5.1 E-Commerce Terminology

Term	Meaning
GMV (Gross Merchandise Value)	Total value of goods sold through the platform before deducting returns, commissions, and discounts. The headline 'size' metric for e-commerce.
Net Revenue / Net Sales	GMV minus returns, cancellations, and seller commissions. The actual revenue recognized by the platform.
Take Rate / Commission Rate	Percentage of GMV that a marketplace earns as commission from sellers. E.g., Flipkart charges 5-15% depending on category.

CAC (Customer Acquisition Cost)	Total marketing/sales spend / Number of new customers acquired. Must be lower than CLV for a sustainable business.
CLV / LTV (Customer Lifetime Value)	Total net profit expected from a customer over their entire relationship with the retailer. Central metric in e-commerce strategy.
Conversion Rate (E-Com)	Percentage of website/app visitors who complete a purchase. Typical e-com conversion: 1–4%.
Bounce Rate	% of users who land on a page and leave without any interaction. High bounce = poor landing page relevance or UX.
Cart Abandonment Rate	% of users who add items to cart but do not complete the purchase. Average is ~70%. Retargeting these users is a major opportunity.
AOV (Average Order Value)	Total Revenue / Number of Orders. E-commerce equivalent of ATV.
ROAS (Return on Ad Spend)	Revenue generated from ads / Ad spend. E.g., ROAS of 4 means every Rs 1 spent on ads generated Rs 4 in revenue.
CPC / CPM	Cost Per Click / Cost Per Thousand Impressions. Key digital advertising metrics.
CTR (Click-Through Rate)	Clicks / Impressions x 100. Measures effectiveness of an ad or email in driving engagement.
SEM / SEO	Search Engine Marketing (paid) / Search Engine Optimization (organic). Both drive traffic to e-commerce sites.
Organic Traffic	Visitors arriving via unpaid channels — SEO, direct, social, email. Indicates brand strength and content quality.
Attribution Model	Rules or algorithms that assign credit for a conversion to the various touchpoints in the customer journey (first click, last click, data-driven).
Fulfillment	The entire process of receiving an order, picking, packing, and shipping it to the customer.
SLA (Service Level Agreement)	Committed delivery timeline. E.g., 'Next Day Delivery' or '2-day SLA'. SLA breach rate is tracked closely.
Return Rate	% of orders returned by customers. High in fashion (30–40%), low in grocery (<5%). A key profitability drag.
NPS (Net Promoter Score)	Customer satisfaction metric. Score = % Promoters - % Detractors. Tracked after delivery and post-return experiences.

5.2 Omnichannel Terminology

Term	Meaning
Omnichannel	Strategy of providing a seamless, integrated customer experience across all channels — physical store, website, mobile app, social commerce.
Click-and-Collect / BOPIS	Buy Online, Pick Up In Store. Customer orders online and collects from the store. Reduces last-mile delivery cost.
Ship-from-Store	Fulfilling online orders by dispatching from the nearest retail store rather than a central warehouse. Speeds up delivery.
ROPO (Research Online, Purchase Offline)	Consumer behavior of researching a product online before buying in a physical store. Accounts for a large share of in-store sales.
Endless Aisle	Using in-store digital kiosks or tablets to allow customers to order items not physically stocked in the store.
Unified Commerce	Next evolution of omnichannel — a single platform managing all sales channels, inventory, and customer data in real time.
Single View of Customer	Consolidating all customer interactions across channels into one unified customer profile. Enables personalization.
Channel Attribution	Identifying which channels (store, app, website, social) contributed to a purchase decision. Complex in omnichannel journeys.
Dark Store	A fulfillment center designed to look like a store but not open to the public. Used to fulfill quick commerce orders.
Micro-Fulfillment Center (MFC)	Small automated fulfillment facilities located close to customers, enabling ultra-fast delivery.

5.3 Common Data Science Use Cases in E-Commerce & Omnichannel

- Recommendation Engine: Suggest relevant products based on browsing history, purchase history, and similar customer behavior.
- Search Ranking / Relevance: Optimize product search results to show the most relevant items first.
- Cart Abandonment Recovery: Identify patterns in cart abandonment and trigger personalized retargeting communications.

- Customer Lifetime Value Prediction: Segment customers by predicted long-term value to allocate acquisition and retention spend.
- Attribution Modeling: Apportion marketing budget across channels based on their contribution to conversion.
- Return Prediction: Predict which orders are likely to be returned to proactively manage fulfillment costs and inventory.

Section 6: Customer Analytics & CRM

Understanding the customer is the ultimate goal of retail data science. Loyalty programs, purchase history, and behavioral data give retailers a rich picture of who their customers are and what they want.

6.1 Customer Analytics Terminology

Term	Meaning
Loyalty Program	Scheme rewarding customers for repeat purchases with points, cashback, or exclusive benefits. Generates invaluable first-party data.
CRM (Customer Relationship Management)	Strategy and technology for managing all customer interactions to maximize retention and value.
RFM Analysis	Recency, Frequency, Monetary analysis. Segments customers by how recently they bought, how often, and how much they spend. Foundation of retail customer segmentation.
Customer Segmentation	Dividing customers into groups with similar characteristics or behaviors to tailor marketing and service.
Churn / Attrition	When a customer stops shopping with a retailer. Churn prediction models identify at-risk customers before they leave.
Reactivation	Marketing efforts to win back lapsed customers who haven't purchased in a defined period.
Share of Wallet	Percentage of a customer's total spending in a category that goes to a specific retailer. Growing share of wallet is a key retention objective.
Customer Lifetime Value (CLV)	Net present value of all future profits from a customer. Guides how much to invest in acquiring and retaining different customer segments.

Customer Acquisition Cost (CAC)	Total cost to acquire one new customer. Must be significantly lower than CLV for the business to be profitable.
First-Party Data	Data collected directly from customers — purchase history, loyalty program data, app usage. Most valuable and privacy-safe.
Third-Party Data	Data purchased from external providers — demographics, lifestyle, location data. Used to enrich customer profiles.
Single View of Customer (SVC)	A unified 360-degree customer record combining online and offline behavior, demographics, and transactions.
Cohort Analysis	Tracking behavior of customers acquired in the same period over time. Reveals retention patterns and how customer quality changes.
Customer Journey	The sequence of touchpoints a customer has with a retailer from awareness to purchase to post-purchase. Journey mapping informs where to invest.
Personalization	Delivering tailored product recommendations, offers, and content to individual customers based on their profile and behavior.
Next Best Action (NBA)	Predicting the most valuable next action to take with a specific customer — which product to recommend, which offer to send, when to contact.

6.2 Common Data Science Use Cases in Customer Analytics

- RFM Segmentation: Classify customers into segments (Champions, Loyal, At Risk, Lost) and design differentiated retention strategies.
- Churn Prediction: Build ML models to identify customers likely to lapse and trigger proactive retention offers.
- CLV Modeling: Predict long-term customer value to guide acquisition spend and tier-based loyalty benefits.
- Personalization Engine: Build collaborative filtering or content-based models to personalize homepage, email, and push notification content.
- Market Basket Analysis: Identify frequently co-purchased products using association rules (Apriori, FP-Growth) to guide bundling and cross-sell.
- Customer Journey Analytics: Map drop-off points in the conversion funnel and prioritize UX improvements.

Section 7: Key Metrics & KPIs You Will Encounter

Retail clients measure performance through a well-defined set of KPIs. Speaking this language fluently will earn immediate credibility in client meetings.

7.1 Sales & Revenue KPIs

KPI	Definition
Total Sales / Net Sales	Top-line revenue. Net Sales = Gross Sales - Returns - Discounts.
Same-Store Sales Growth (SSG) / Comp Growth	Revenue growth from stores open 12+ months. Strips out new store openings. Core organic growth metric.
Sales per Square Foot	Net Sales / Total Selling Area. Standard measure of store productivity.
Sales per Employee	Net Sales / Number of Employees. Measures workforce productivity.
Conversion Rate	Transactions / Footfall x 100. % of visitors who make a purchase.
ATV (Average Transaction Value)	Total Sales / Number of Transactions. Average spend per customer visit.
UPT (Units per Transaction)	Total Units Sold / Number of Transactions. Items per basket.
Footfall / Traffic	Number of customers entering the store. Drives all other performance metrics.
Revenue per Visitor (RPV)	Net Revenue / Unique Visitors. E-commerce metric combining conversion and AOV.
GMV (Gross Merchandise Value)	Total value of orders before deductions. Primary size metric for e-commerce and marketplaces.

7.2 Profitability KPIs

KPI	Definition
Gross Margin (%)	(Net Sales - Cost of Goods Sold) / Net Sales x 100. Profitability after direct product costs.
Gross Profit	Net Sales - COGS. Absolute profit before operating expenses.
EBITDA	Earnings Before Interest, Tax, Depreciation and Amortization. Operational profitability measure.

GMROI	Gross Margin / Average Inventory at Cost. Profit earned per rupee invested in inventory.
Contribution Margin	Revenue - Variable Costs. Useful for evaluating individual promotions, categories, or channels.
Shrinkage as % of Sales	Inventory loss / Net Sales. Typically 1–3%. Directly reduces gross margin.
Operating Cost Ratio	Operating Expenses / Net Sales. Lower is better. Retail efficiency benchmark.
EOSS Margin Impact	Gross margin dilution caused by end-of-season sale markdowns. Tracked carefully by merchandise teams.

7.3 Inventory KPIs

KPI	Definition
Sell-Through Rate (%)	Units Sold / Units Received x 100. % of stock sold. Fashion target: 70%+ before markdown.
Weeks of Supply (WOS)	On-Hand Inventory / Average Weekly Sales. How many weeks stock will last.
Inventory Turnover	Net Sales or COGS / Average Inventory. How efficiently inventory is being converted to sales.
In-Stock Rate / OSA (%)	% of time all ranged products are available on shelf. Target: 95%+.
Stockout Rate (%)	% of SKUs out of stock at any given time. Direct driver of lost sales.
Days on Hand (DOH)	Average days of inventory on hand. Lower = leaner. But too low risks stockouts.
Markdown Rate (%)	Value of markdowns / Net Sales. High markdown rate signals poor buying decisions or over-buying.
Obsolete / Aged Stock (%)	Stock older than defined age threshold / Total inventory. Tied-up capital needing action.

7.4 Customer & Loyalty KPIs

KPI	Definition

Active Customer Base	Number of customers who made at least one purchase in the last 12 months.
Retention Rate (%)	% of customers from one period who purchase again in the next period.
Churn Rate (%)	% of customers who do not return after a defined period of inactivity.
Average Purchase Frequency	Number of transactions per customer per year. Key driver of sales growth alongside spend per visit.
CLV (Customer Lifetime Value)	Predicted total net profit from a customer over their relationship with the retailer.
Loyalty Program Penetration (%)	% of transactions linked to a loyalty program member. Higher = more first-party data.
NPS (Net Promoter Score)	% Promoters - % Detractors. Overall satisfaction and advocacy metric.
Return Rate (%)	Returns / Gross Sales. High in fashion; a key profitability concern in e-commerce.

Section 8: Data Landscape in Retail

8.1 Types of Data You Will Work With

- POS (Point of Sale) Data: Transaction-level data — item, quantity, price, discount, store, timestamp, cashier. The most granular and voluminous data source in retail.
- Loyalty / CRM Data: Customer-level data linked to purchases via loyalty card or app. Enables RFM analysis and personalization.
- Footfall Data: Store visitor counts from footfall counters (infrared, thermal, video). Used with POS data to calculate conversion rate.
- Inventory / Stock Data: Current stock levels at each store and DC, purchase orders, GRNs, stockouts, shrinkage.
- Supplier / Vendor Data: Lead times, fill rates, pricing, product catalogs. Used in buying and supply chain analytics.
- Web & App Behavioral Data: Clickstreams, search queries, product views, add-to-cart events, session duration. Rich for e-commerce projects.
- Marketing Data: Campaign spends, impressions, clicks, response rates across digital and offline channels.
- Competitor / Market Data: Competitor prices (scraped), market share data (Nielsen), consumer research (panel data).
- Weather & Calendar Data: External factors that affect retail demand — temperature, rain, public holidays, school calendar, festival dates.

- Spatial / Location Data: Store catchment demographics, footfall heatmaps, competitive proximity. Used in site selection and cluster analysis.

8.2 Key Systems & Technologies

System / Tool	What It Is
POS System	Point of Sale terminal software that processes transactions and generates sales data. E.g., NCR, Oracle MICROS, Ginesys (India), LS Retail.
ERP (Enterprise Resource Planning)	Central system managing financials, inventory, procurement, and HR. SAP S/4HANA and Microsoft Dynamics are common in retail.
WMS (Warehouse Management System)	Manages all operations within a warehouse — receiving, storage, picking, packing, and dispatch. E.g., Manhattan Associates, Blue Yonder.
OMS (Order Management System)	Manages order lifecycle across channels — captures, routes, and tracks orders from placement to delivery. Critical for omnichannel.
CRM System	Customer data and relationship management platform. E.g., Salesforce, Adobe Experience Cloud, SAP CX.
CDP (Customer Data Platform)	Unified platform that consolidates first-party customer data from all touchpoints into a single customer profile. Enables personalization at scale.
E-Commerce Platform	Technology powering the online store. E.g., Shopify, Magento, SAP Commerce Cloud, custom-built platforms.
BI / Analytics Tool	Dashboarding and reporting. Power BI, Tableau, Looker are standard. Many retailers build custom analytics layers on top.
Demand Planning Tool	Forecasting and replenishment software. E.g., Blue Yonder (JDA), o9, Relex, SAP IBP.
Loyalty Platform	Manages loyalty program enrollment, points, redemptions, and tier management. E.g., Capillary Technologies (widely used in India), Comarch.
Price Intelligence Tool	Monitors competitor pricing in real time via web scraping. E.g., Wiser, Intelligence Node, Price2Spy.
Personalization Engine	Delivers personalized recommendations and content. E.g., Adobe Target, Dynamic Yield, Insider, in-house ML models.

Section 9: Quick Reference Glossary — A to Z

A rapid-fire glossary of retail terms commonly used in client meetings, project briefs, and dashboards.

Term / Abbreviation	Quick Definition
Above the Fold	Content visible on a webpage without scrolling. Prime real estate for e-commerce banners and promotions.
Adjacencies	Products placed next to each other in a store to encourage cross-category purchase. E.g., pasta adjacent to pasta sauce.
A/B Testing	Controlled experiment comparing two versions (A and B) of a webpage, email, or promotion to determine which performs better.
BOPIS	Buy Online, Pick Up In Store. A popular omnichannel fulfillment option.
Breadcrumb	Navigation trail on a website showing the user's path. Also a UX term for helping users navigate back.
Chargeback	A reversal of a credit card transaction initiated by the customer's bank after a dispute. Retail and e-commerce fraud risk.
COD (Cash on Delivery)	Payment collected at delivery. Common in India. Higher return rates and fraud risk than prepaid orders.
COGS (Cost of Goods Sold)	Direct cost of purchasing the products sold. Gross Margin = Net Sales - COGS.
Collaborative Filtering	Recommendation algorithm that suggests products based on behavior of similar users. Powers most e-com recommendation engines.
Dead Zone	Area of a store that receives very low customer traffic and therefore poor sales. Redesigning dead zones is a store analytics project.
Endless Aisle	In-store kiosks allowing customers to browse and order products not physically available in that store.
EOSS (End of Season Sale)	Major markdown event at end of fashion season to clear remaining stock.

Facing	The number of product units visible at the front of a shelf. More facings = more visibility = higher sales.
FBA (Fulfillment by Amazon)	Amazon fulfills orders on behalf of third-party sellers using its own warehouses and logistics.
First-Party Cookie	Data collected directly by a website about its own visitors. More privacy-safe than third-party cookies.
Footprint	The total number of stores a retailer operates. Expanding or rationalizing footprint is a strategic decision driven by analytics.
Ghost Cart	Shopping cart with items added but never checked out. The most common e-commerce abandonment scenario.
Heat Map	Visual representation of where customers move or click most in a store or on a website. Used in layout and UX optimization.
Hyper-Personalization	Using real-time data and AI to deliver one-to-one tailored experiences to every individual customer.
In-Store Analytics	Use of cameras, sensors, and Wi-Fi to understand customer movement, dwell time, and engagement within physical stores.
Joint Business Plan (JBP)	Annual plan agreed between a retailer and a key supplier on sales targets, promotions, and shared investments.
KVI (Key Value Item)	A product that consumers use as a price benchmark when evaluating a retailer's overall value. Retailers price KVIs very competitively.
Last Mile	Final leg of delivery from local hub to customer. The most expensive part of e-commerce fulfillment.
Listing	A product page on an e-commerce marketplace. Listing quality (images, descriptions, keywords) directly affects conversion.
Margin Mix	How the blend of high- and low-margin products in the overall sales mix affects total profitability.
Mystery Shopping	Practice of sending undercover shoppers to evaluate store experience, staff behavior, and compliance.
NPS (Net Promoter Score)	% Promoters - % Detractors. Standard customer satisfaction metric across retail.
OOS (Out of Stock)	A product that is unavailable when a customer wants to buy it. One of the biggest causes of lost sales in retail.

Phantom Inventory	System showing stock as available when it is physically missing, damaged, or misplaced. Causes ghost stockouts.
Pre-Season Planning	Buying and assortment decisions made before a season starts, based on trend forecasts and historical data.
Product Detail Page (PDP)	The individual product page on an e-commerce site. Conversion rates on PDPs are a key optimization metric.
RFID (Radio Frequency Identification)	Technology using tags to track inventory location in real time, improving stock accuracy and replenishment speed.
SKU Velocity	Speed at which a SKU sells. High-velocity SKUs need frequent replenishment; low-velocity SKUs are markdown candidates.
Slotting Fee	Fee paid by suppliers to retailers for premium shelf placement or to list a new product.
Social Commerce	Selling products directly through social media platforms (Instagram Shopping, WhatsApp Commerce, Pinterest).
Stock Cover	Number of days or weeks of sales covered by current on-hand inventory. Similar to WOS.
Top-of-Funnel	Early stages of customer journey — awareness and discovery. Typically driven by brand advertising.
Trading Density	Net Sales / Selling Area (sq ft or sq m). Measures how productively space is being used to generate revenue.
Traffic Driver	A product or promotion that brings customers into the store or onto the website, even if the item itself has low margin.
VOC (Voice of Customer)	Feedback collected from customers through surveys, reviews, and social listening to understand satisfaction and unmet needs.
Whitespace	Categories or geographies where a retailer has no presence but should, based on customer demand or competitive gaps.

Section 10: Tips for Working with Retail Clients

Do's for Data Scientists in Retail

- Always start by understanding the channel mix — physical store vs. e-commerce vs. omnichannel. The data, KPIs, and problems are very different.

- POS data is your best friend. Learn its structure deeply — store, SKU, timestamp, quantity, price, discount. Most retail DS problems start here.
- Talk in business terms: 'This model will reduce stockouts by X%, recovering INR Y crore in lost sales annually' beats 'accuracy improved by 5%'.
- Understand the promotional calendar before building any demand forecast — promotions and seasonal events dominate retail demand patterns.
- Ask about loyalty program penetration early. If only 30% of transactions are linked to a loyalty ID, your customer analytics will have significant gaps.
- Segment everything by store cluster and channel. National averages in retail are misleading — a top-performing Tier 1 metro store is very different from a Tier 3 town store.
- Validate inventory data quality before any supply chain project. Phantom inventory and stockroom vs. shelf mismatches are common.

Watch Out For

- Holiday / Seasonal Spikes: Christmas, Diwali, Big Billion Day, EOSS — always model these explicitly or your forecasts will be off.
- Phantom Inventory: System may show stock as available when it is actually missing or damaged. This causes ghost stockouts in models.
- POS Data Gaps: Not all transactions may be captured — cash transactions, staff purchases, damage write-offs can create holes in the data.
- Return Contamination: Returned items re-enter inventory and can distort sell-through and turnover calculations. Always check your return handling logic.
- Omnichannel Attribution: A customer may browse online and buy in store. Last-click attribution will under-credit digital channels.
- Price Change History: Price changes must be tracked as a dimension in demand forecasting models. If a price drop happened 3 months ago, your historical baseline is contaminated.
- Store Opening / Closing Events: New stores ramp up, closing stores ramp down. These must be excluded from comparable store analyses.

You are now equipped with the foundational knowledge to confidently navigate Retail client engagements. Good luck!